UNICORN VALVES

Revolutionizing Flow Control

PRODUCT BROCHURE

CONTROL VALVES & ACTUATORS



ISO 9001 : 2015
Quality Management
System Certification

CONTROL VALVES



91000 SERIES
TOP GUIDING GLOBE

Sizes: 0.5" - 8"

Pressure Rating and End Connections:
Flanged - ASME 150~ASME 2500
and DIN PN 6 ~ PN 250
Welded- BW and SW
Threaded- NPT 0.5"~2"

Bonnet Options:
Standard / Bellow Sealed
Extended / Cryogenic

Characteristics:
Linear / Equal Percentage / Quick
Opening

Leakage Class:
Standard: As per ANSI/FCI 70-2 IV/V
Option: As per ANSI/FCI 70-2 VI

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
Contoured Unbalanced Plug
Deci-Nill® and Cav-Kill®
Single stage / Double stages
Steam jacket / Vacuum service
Soft seat / Low emission packing
Expander with cartridge for reduction
of very high noise

92000 SERIES
TOP GUIDING ANGLE

Sizes: 0.5" - 8"

Pressure Rating and End Connections:
Flanged - ASME 150 - ASME 2500 and
DIN PN 6 ~ PN 250
Welded - BW & SW
Threaded - NPT 0.5" ~ 2"

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear / Equal Percentage
Quick Opening

Leakage Class:
Standard: As per ANSI/FCI 70-2 IV/V
Option: As per ANSI/FCI 70-2 VI

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
Contoured Unbalanced Plug
Deci-Nill® and Cav-Kill®
Single stage / Double stages
Steam jacket / Vacuum service
Soft seat / Low emission packing
Expander with cartridge for reduction
of very high noise

93000 SERIES
MICRO FLOW GLOBE

Sizes: 0.5", 0.75" - 1"

Pressure Rating and End Connections:
Flanged - ASME 150 - ASME 2500
and DIN PN 6 ~ PN 250
Welded - BW & SW
Threaded - NPT 0.5" ~ 1"

Bonnet Options:
Standard /Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear / Equal Percentage

Leakage Class:
Standard: As per ANSI/FCI 70-2 V
Option: As per ANSI/FCI 70-2 VI

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
Contoured Micro flow control
Solid Stellite Single Piece Plug & Stem
Deci-Nill® and Cav-Kill®
Single stage / Double stages
Steam jacket / Vacuum service
Soft seat / Low emission packing
Expander with cartridge for reduction
of very high noise

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94000 SERIES
CAGE GUIDING GLOBE

Sizes: 2" - 36"

Pressure Rating and End Connections:
Flanged - ASME 150 - ASME 2500
and DIN PN 6 ~ PN 250
Welded - BW & SW
Threaded - NPT 0.5" ~ 2"

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear / Equal Percentage
Quick Opening / Modified Equal Percentage

Leakage Class:
Standard: As per ANSI/FCI 70-2 IV
Option: As per ANSI/FCI 70-2 V

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/ Double Acting)
Electrical Actuator

Other Features:
Pressure Energised Seal Ring
Auxilliary Pilot Plug Design
Deci-Nill® and Cav-Kill® / Severe Service
Application / Diffuser seat ring design
Single stage/Double stage/ Three stage design
Steam jacket / Vacuum service
Expander with cartridge for reduction of
very high noise / Low emission packing



95000 SERIES
CAGE GUIDED ANGLE

Sizes: 2" - 24"

Pressure Rating and End Connections:
Flanged - ASME 150 - ASME 2500 and
DIN PN 6 ~ PN 250
Welded - BW & SW
Threaded - NPT 0.5" ~ 2"

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear /Equal Percentage
Quick Opening / Modified Equal Percentage

Leakage Class:
Standard: As per ANSI/FCI 70-2 IV
Option: As per ANSI/FCI 70-2 V

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/ Double Acting)
Electrical Actuator

Other Features:
Pressure Energised Seal Ring
Auxilliary Pilot Plug Design
Deci-Nill® and Cav-Kill® / Severe Service
Application / Diffuser seat ring design
Single stage/Double stage/ Three stage design
Steam jacket / Vacuum service
Expander with cartridge for reduction of
very high noise / Low emission packing



96000 SERIES
3 WAY MIXING / DIVERTING

Sizes: 0.75" - 24"

Pressure Rating and End Connections:
Flanged - ASME 150 - ASME 2500 and
DIN PN 6 ~ PN 250
Welded - BW & SW
Threaded - NPT 0.5" ~ 2"

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear / Equal Percentage
Quick Opening / Modified Equal Percentage

Leakage Class:
Standard: As per ANSI/FCI 70-2 IV
Optional: As per ANSI/FCI 70-2 VI

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

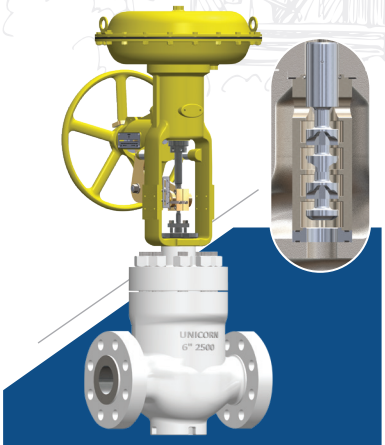
Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/ Double Acting)
Electrical Actuator

Other Features:
HVAC Application
Temperature Control Services
Flow Diverting Applications
Steam Jacket / Low Emission packing



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CRITICAL SERVICE VALVES



97000 SERIES
MULTI STEP MULTI STAGE AXIAL FLOW

Size: 1" - 8"

Pressure Rating and End Connections:
Flanged - ASME 150~ASME 2500
and DIN PN 6 ~ PN 250
Welded- BW and SW

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

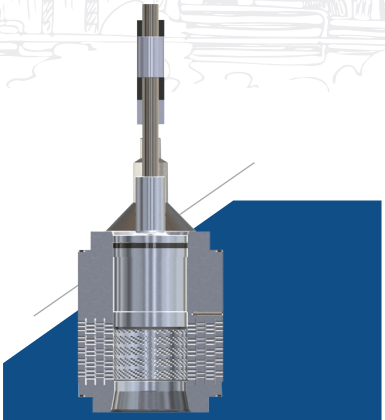
Characteristics:
Modified Equal Percentage

Leakage Class:
Standard: Class IV / V
Optional: ANSI /FCI 70-2 VI

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
High Pressure Drop Appllications
Velocity Control Trim
Low Emission Packing
Cavitation Control (Cav-Kill®) / Noise Reduction
(Deci-Nill®) Trim / Multistep (Axial) Flow Trim



98000 SERIES
MULTI STAGE RADIAL FLOW

Size: 2" - 24"

Pressure Rating and End Connections:
Flanged - ASME 150~ASME 2500
and DIN PN 6 ~ PN 250
Welded- BW and SW

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

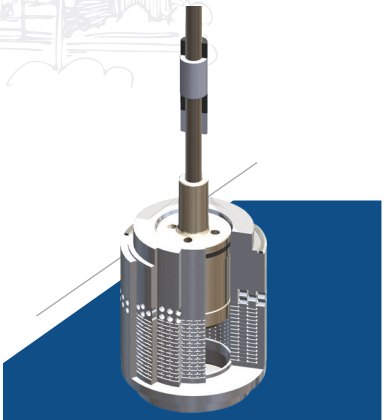
Characteristics:
Linear / Equal Percentage
Modified Linear & Equal Percentage

Leakage Class:
Standard: Class IV
Optional: ANSI /FCI 70-2 V

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
Low Emission Packing
Cavitation Control (Cav-Kill®) /
Multi - Stage Multi - Cage (Radial) Flow Trim
Velocity Control Trim



99000 SERIES
MULTI STAGE AXIAL FLOW

Size: 2" - 24"

Pressure Rating and End Connections:
Flanged - ASME 150~ASME 2500
and DIN PN 6 ~ PN 250
Welded - BW and SW

Bonnet Options:
Standard / Extended
Bellow Sealed / Cryogenic

Characteristics:
Linear / Equal Percentage
Modified Linear & Equal
Percentage

Leakage Class:
Standard: Class IV
Optional: ANSI /FCI 70-2 V

Actuators:
Pneumatic Spring Diaphragm
(Single Acting)
Pneumatic Piston Cylinder
(Single/Double Acting)
Electrical Actuators

Body Material:
Carbon Steel
Alloy Steel
Stainless Steel
Duplex Stainless Steel
Super Duplex Stainless Steel
Exotic Materials

Other Features:
Low Emission Packing
Cavitation Control (Cav-Kill®) /
Multi - Stage Multi - Cage (Axial) Flow Trim
Velocity Control Trim



ACTUATORS

ACTUATORS

UA - 11/12 SERIES
PNEUMATIC SPRING DIAPHRAGM TYPE



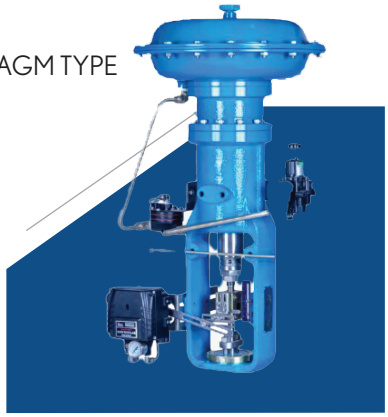
With Handwheel

Type:
Pneumatic Spring
Diaphragm - Single Acting Type

Model Nos.:
UA - 11 (Direct Acting)
UA - 12 (Reverse Acting)

Sizes:
30, 35, 40, 45, 50

Design Pressure:
Upto 5 Bar G



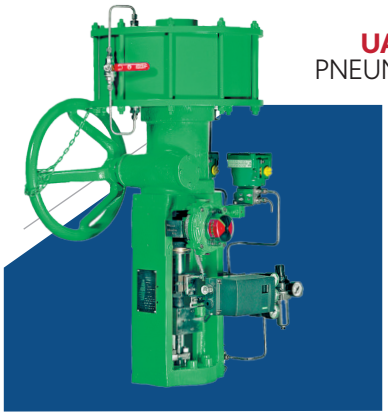
Without Handwheel

Spring Range:
3-15 psi, 6-30 psi
12-30 psi

Design Pressure:
19.05mm - 101.6mm

Other Features:
Manual Override
Position Indicator
Adjustable Mechanical Stopper

UA - 30/31/32 SERIES
PNEUMATIC PISTON CYLINDER

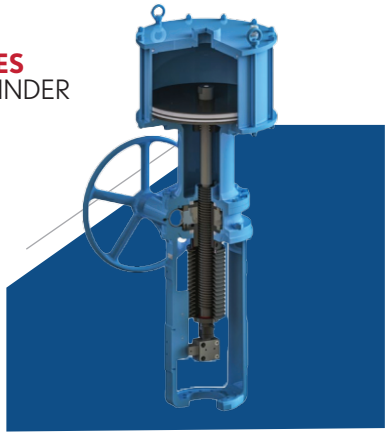


Slide Mounted Handwheel

Type:
Piston Cylinder - Double Acting Type
Piston Cylinder - Single Acting Type

Model Nos.:
UA - 30 (Double Acting)
UA - 31 (Single Acting Direct)
UA - 32 (Single Acting Reverse)

Sizes:
7.5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60



Cut Section View

Stroke:
15mm - 250mm

Thrust:
100,000 N

Design Pressure:
Up to 10 Bar G

Other Features :
Manual Override
Position Indicator
Backup System for achieving fail action

ACTUATORS

UNICORN VALVES
THE VALVE ENGINEERING EXPERTS



UNICORN VALVES
Revolutionizing Flow Control

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